



YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRICAL ENGINEERING
Project

on

**Advanced Monitoring and Control Mechanism
for LVDC Microgrid**

Presentation by

(1) Devanshu Nishibkar

(2) Viraj Gaikwad

(3) Pragati Wange

(4) Aniket Naranje

(VII SEM B.E. Electrical Engineering)

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Under the guidance of

Prof. P. S. Shete
(Assistant Professor)

Er. Suyash Aglave
(Nayak Power systems Pvt.Ltd,Thane)

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Aim :

To design and develop a Low Voltage DC (LVDC) microgrid prototype that can monitor, control, and protect the system in real time using an ESP32-based intelligent control unit with automatic switching between grid and backup sources.

Objectives :

- Monitoring and mitigation in the LVDC microgrid.
- To design a 120 V DC bus using grid supply (via SMPS) and a battery backup system.
- To ensure system safety under varying load conditions.

Introduction

- The growing demand for efficient, flexible, and reliable power systems has led to the increasing adoption of LVDC microgrids.
- This project focuses on developing a prototype LVDC microgrid that can operate in both grid-connected and standalone (UPS) modes.
- An ESP32 microcontroller acts as the central unit, controlling sugar cube relays for source switching and overload protection, while continuously monitoring voltage and current.
- The system displays data on an LCD and through an IoT-based dashboard, ensuring advanced monitoring and predictive control of the DC bus.

❖ **Key Aspects of the hardware:**

1. Designed a 120 V DC bus using grid (SMPS) and battery (inverter) sources.
2. Used sugar cube relays for source switching and overload tripping.
3. Implemented ESP32-based intelligent control for automation and protection.
4. Employed voltage divider and current sensing circuits for real-time monitoring.
5. Ensured system protection through real-time current sensing logic.

Literature Review

Sr. No.	Title of Paper	Author Names	Year	Finding	Use for work Research work
1	Practical Implementation of a Public LVDC Microgrid: Challenges and Experiences	Tekai Eddine Khalil Zidane, Zarina Ab Muis, Wai Shin Ho, Younes Zahraoui, Ali Saleh Aziz, Ward Ysebie, Hakim Azaïoud, Brecht Caers, Jan Desmet	2025	The core findings of the paper are that off-grid LVDC solutions significantly boost energy efficiency (up to 30%) and reliability, but grid-connected systems remain the most economical. Crucially, the biggest barrier to widespread public LVDC microgrid implementation is the lack of a harmonized global standard for voltage levels and protection systems, which complicates design and delays adoption. Practical experience confirms the potential of DC technology for integrating renewables but highlights the major technical challenge of designing DC protection mechanisms without a natural current zero-crossing.	The key finding is that while LVDC boosts efficiency, the widespread practical implementation is severely hindered by the lack of standardized DC voltage levels and reliable DC protection systems..
2	A Cooperative Power Flow Control Strategy for Low-Voltage DC Microgrid Clusters	Hongqin Yin, Fei Gao, Muhammad Mansoor Khan, and Dong Liu	2025	The paper's findings center on a Cooperative Power Flow Control Strategy designed for LVDC microgrid clusters to achieve economically optimal operation and reliable power sharing. It utilizes a hierarchical control structure with a tertiary layer employing a consensus-based algorithm to minimize overall generation costs and losses across the cluster. Ultimately, this approach successfully ensures accurate DC voltage regulation and efficient power management through low-bandwidth communication, enhancing the system's scalability and resilience.	accurate DC voltage regulation across interconnected LVDC microgrid clusters.
3	Self-sufficiency and lifetime improvement of community BESS on an LVDC backbone compared to individual BESS	Azaïoud Hakim, J. Knockaert, Lieven Vandevelde, Jan Desmet	2023	The core findings are that implementing a community Battery Energy Storage System (BESS) on an LVDC backbone offers superior performance compared to individual household BESS units. This communal approach significantly improves community self-sufficiency and self-consumption ratios, while also allowing for a substantial reduction in the overall BESS capacity (35% to 68% smaller) needed to meet the same self-sufficiency targets. Furthermore, the centralized BESS setup enhances efficiency by reducing DC/DC conversion losses and offers a slight improvement in the battery's overall lifetime	Implement the cooperative control to allow seamless power transfer between microgrid units during disturbances, thereby significantly improving service continuity

Methodology

- Studied the architecture and power flow of the LVDC microgrid system.
- Selected 120 V DC as the bus voltage for the prototype .
- Connected a 12 V, 8 Ah battery with an inverter circuit to supply 120 V DC during grid failure.
- Implemented ESP32 microcontroller for real-time monitoring and control.
- Used voltage divider and current sensing circuits for continuous measurement of bus parameters.
- Controlled sugar cube relays for automatic source switching and overload protection under varying load conditions.

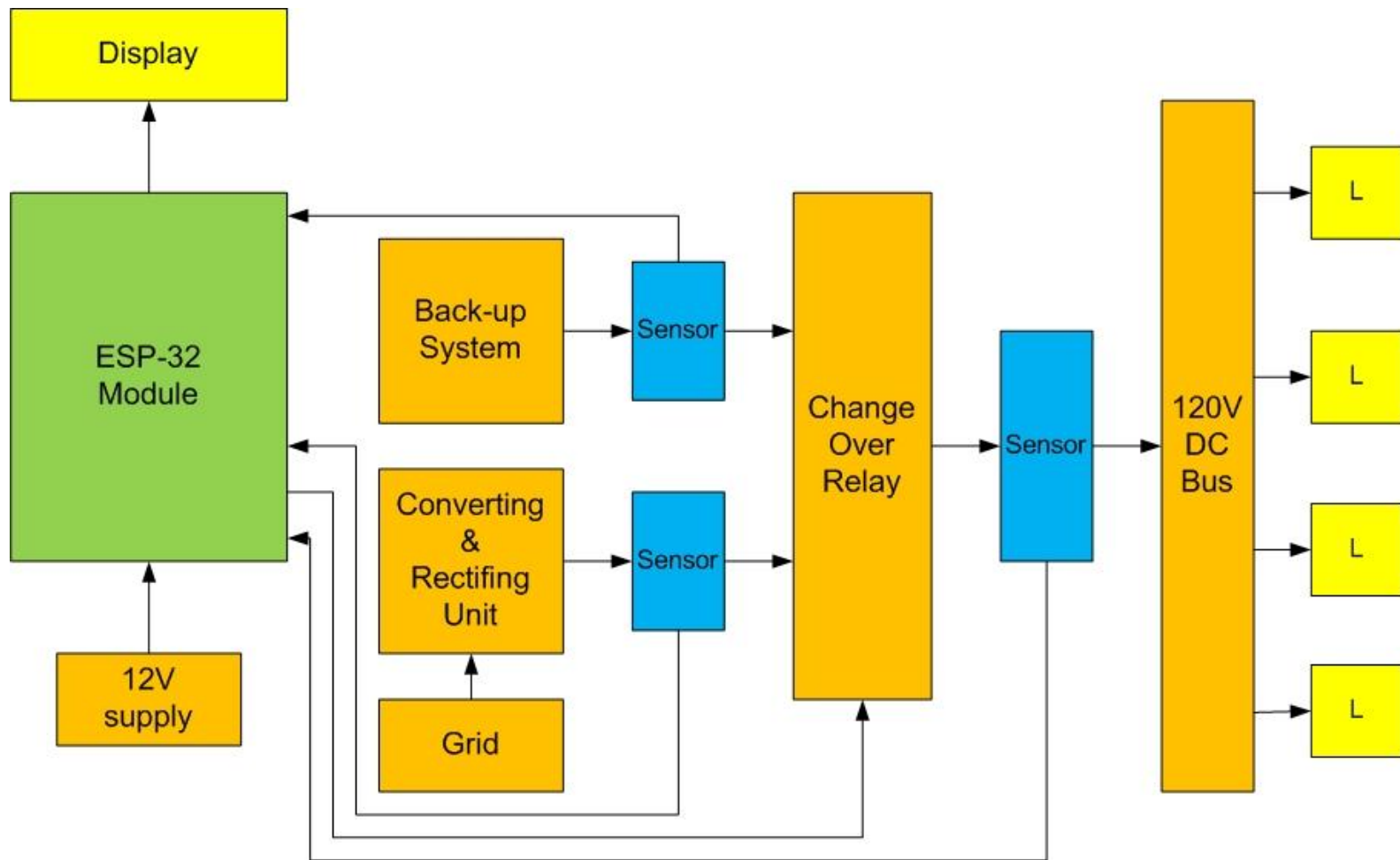


Fig.1: Block Diagram Prototype for LVDC Microgrid

Components Details

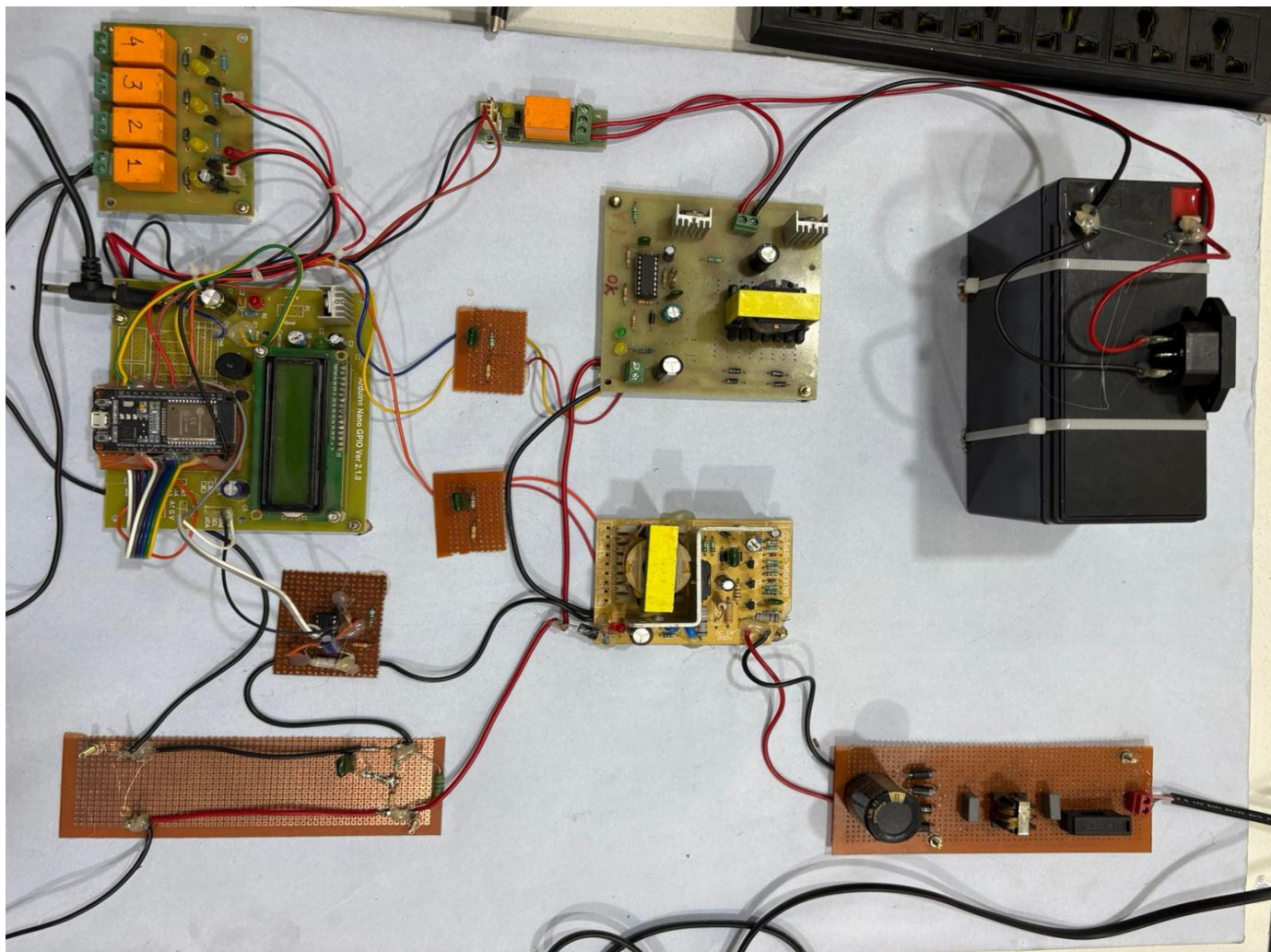


Fig.2 : Modelled Hardware

- **SMPS**

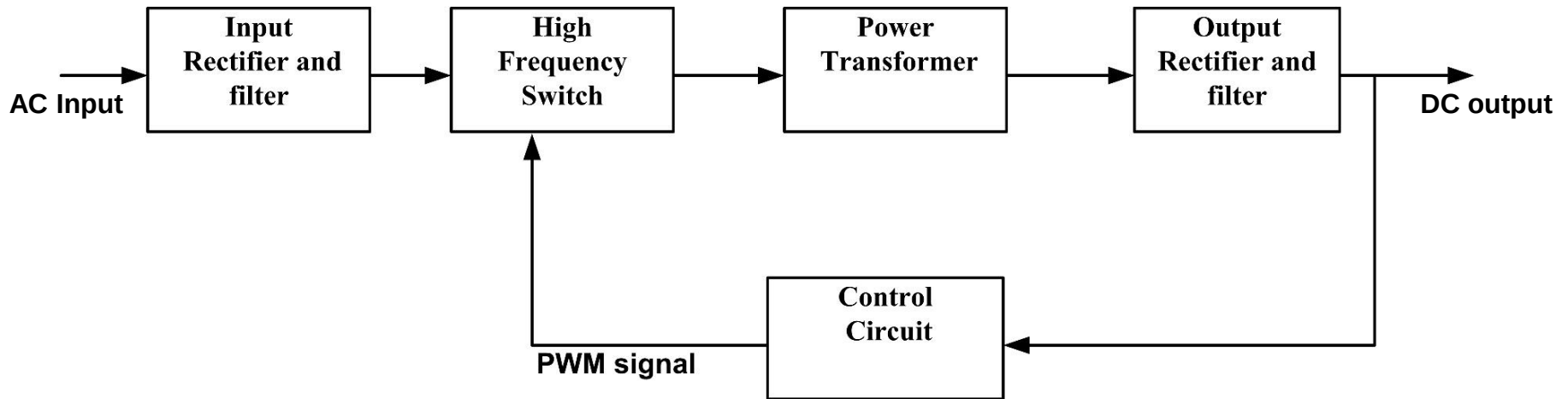
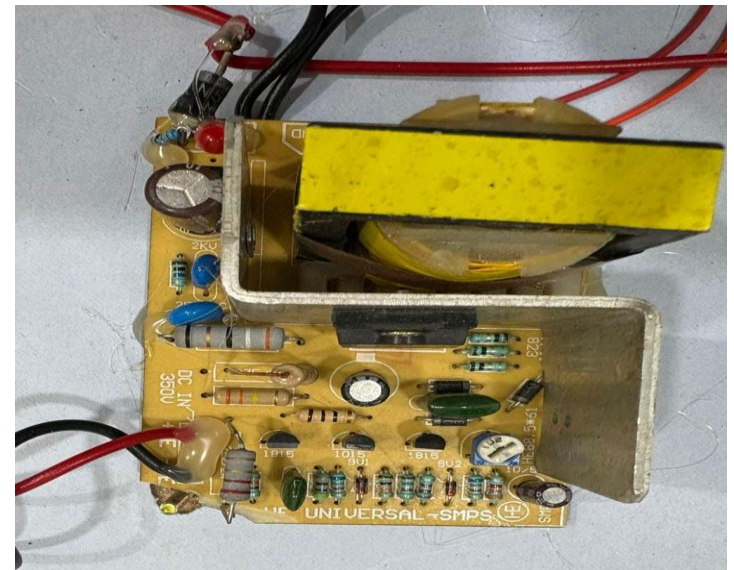
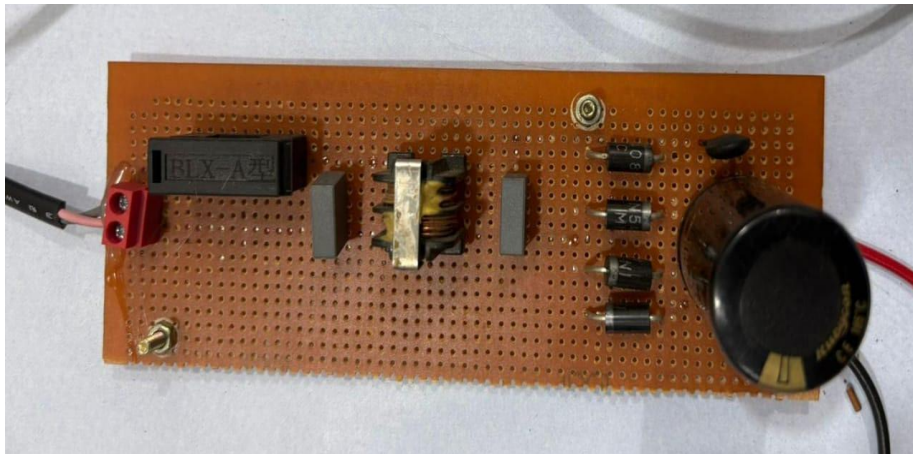


Fig.2: SMPS Block diagram



• Input Rectifier and Filter

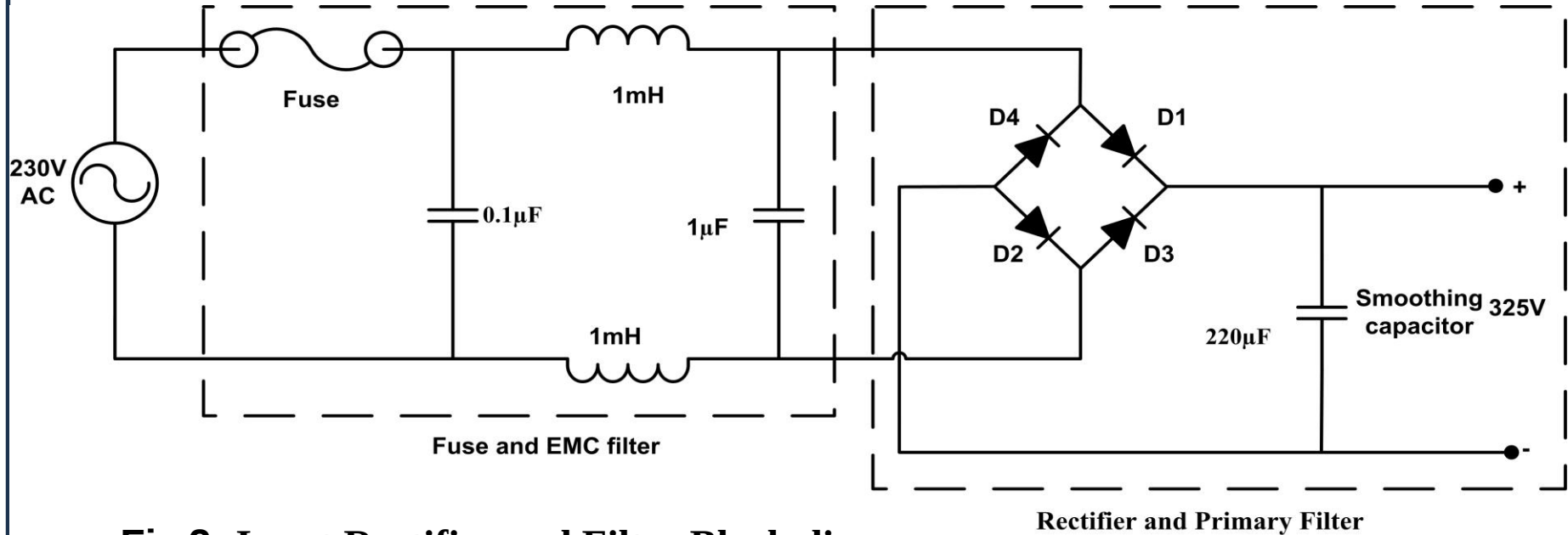
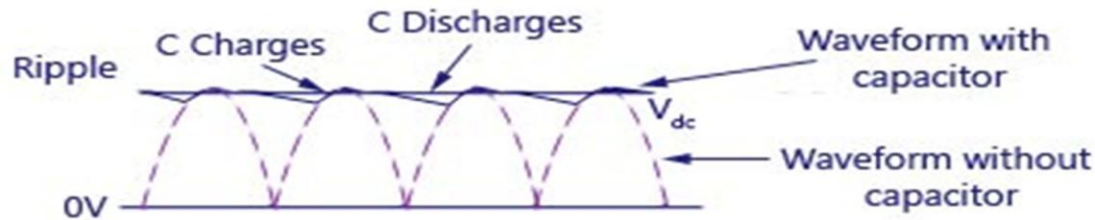


Fig.2: Input Rectifier and Filter Block diagram

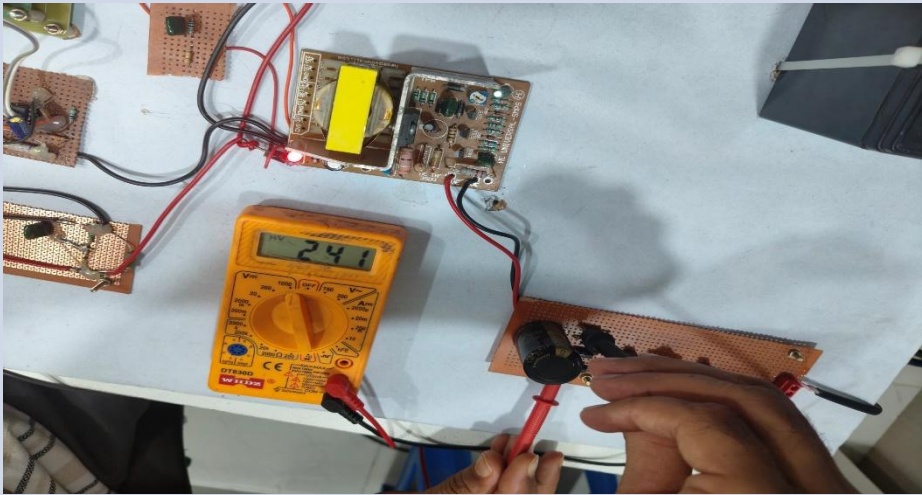
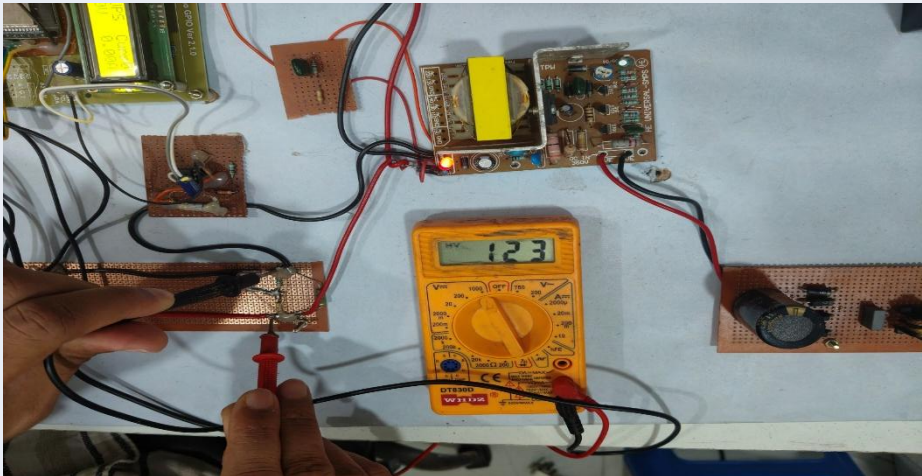


- It converts 230V AC to 325 V DC.
- The EMC filter is used to remove higher frequencies.
- Smoothing Capacitor help us to reduce ripple and provide steady DC output voltage

- **Self Oscillating SMPS to convert 325V DC to 120V DC**

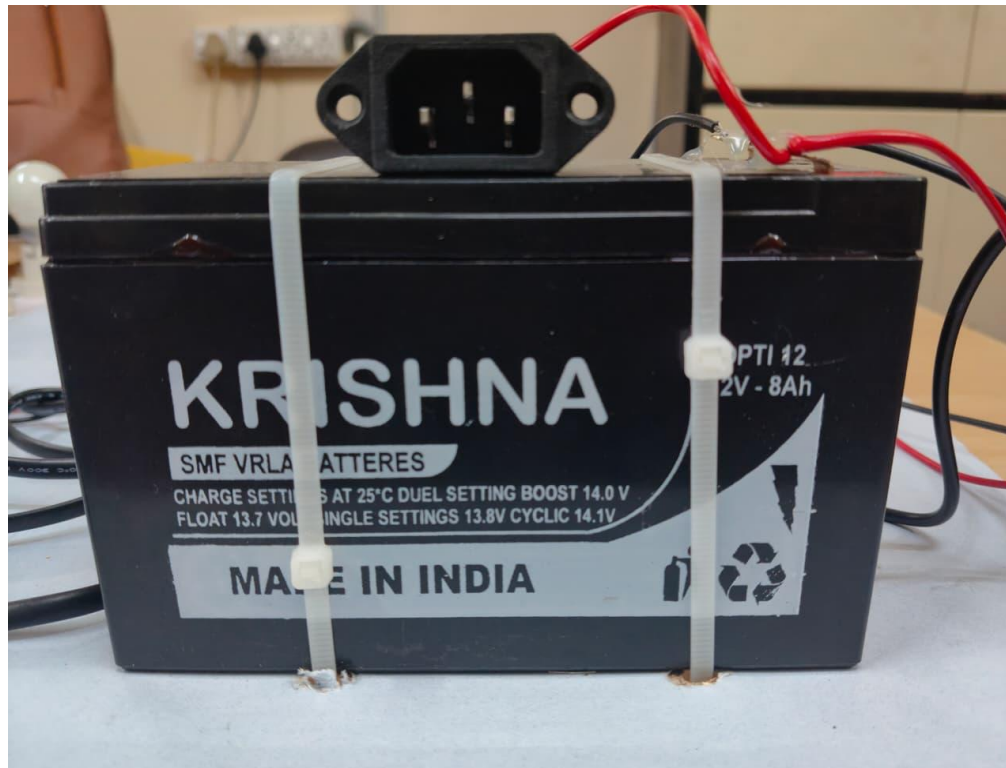
- Input (325V DC): The raw, high-voltage DC is fed into the circuit.
- Switching Action: The Transistor is driven by Positive Feedback from the Auxiliary Winding to continuously turn ON and OFF. This chops the 325V DC into high-frequency pulses.
- Transformer: When the switch is ON, the transformer stores energy. When the switch is OFF, the transformer transfers energy to the secondary side.
- Output (120V DC): The secondary winding's pulses are rectified and smoothed using capacitor to get the desired 120V DC output .



Sr.no.	Location	Measurement
1	Rectifier o/p	
2	DC Bus Voltage	

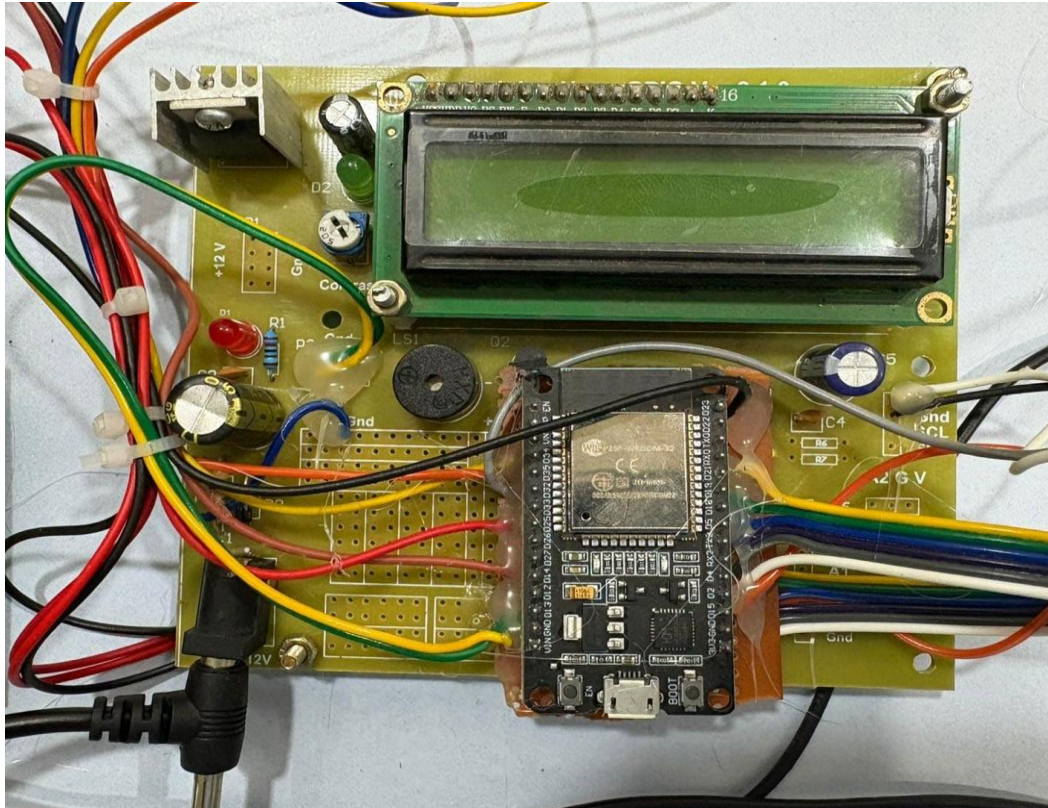
Battery CP12120:

- The Component is useful for the Back-up in the System.
- The Rating of battery is 12V, 8AH .
- It is Maintenance-Free Rechargeable Sealed Lead-Acid (SLA) Battery.



ESP-32 Module with Display:

- This Module Contain ESP-32 microcontroller with 30 pins, Display 16x2 LCD, Buzzer, 12V Supply and Converting circuit.
- This Module help to control and monitor the whole Hardware System.
- The ESP takes Analog input from the Sensors and convert it into Digital to show on display LCD and in Mobile using via Wi-Fi.



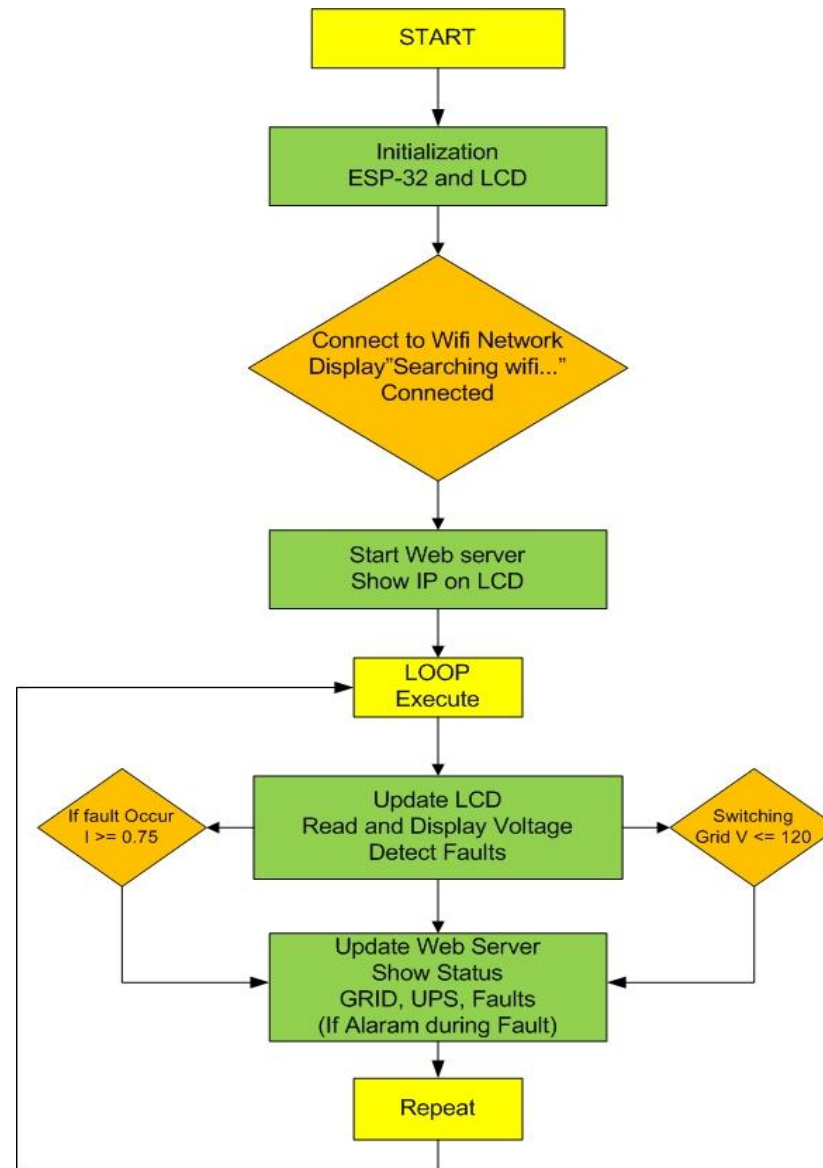
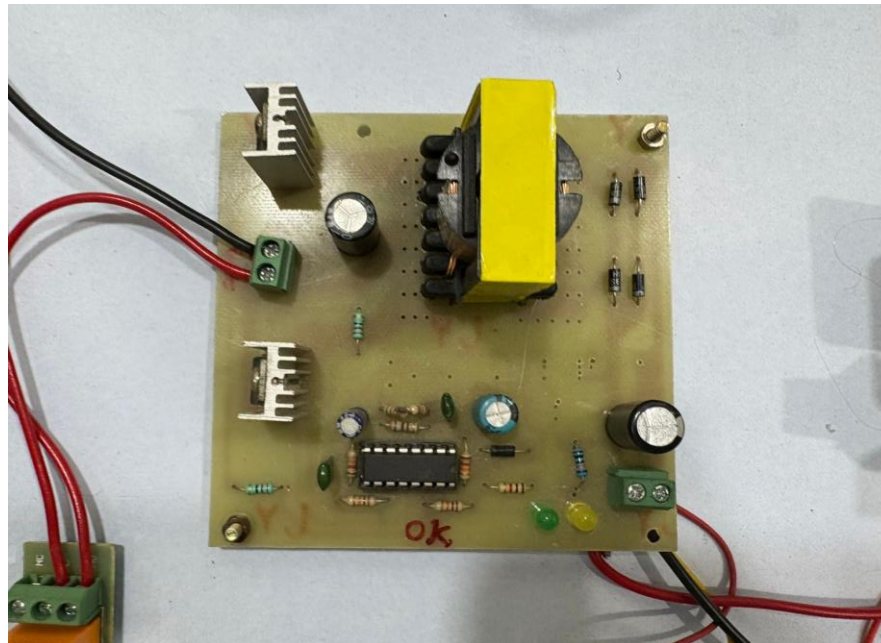


Fig.3 : Flowchart of program in ESP-32 Microcontroller

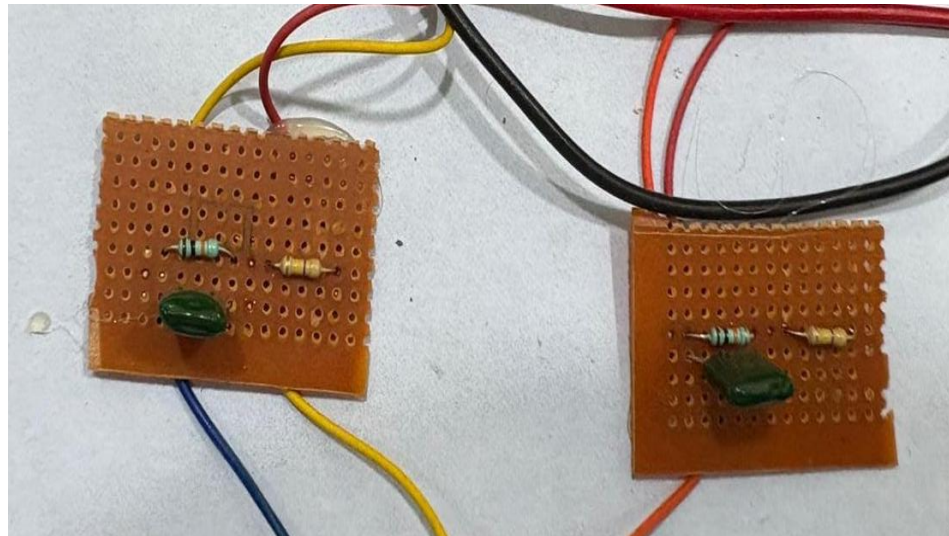
- **Inverter module**

- The Black IC Chip acts as the PWM Controller. It generates the precise, high-frequency square wave signals required to ensure efficient and controlled power switching.
- The Electrolytic Capacitors are used for filtering and energy storage. They stabilize the input voltage and smooth the high-frequency switching noise, ensuring the control IC and the power stage operate reliably.



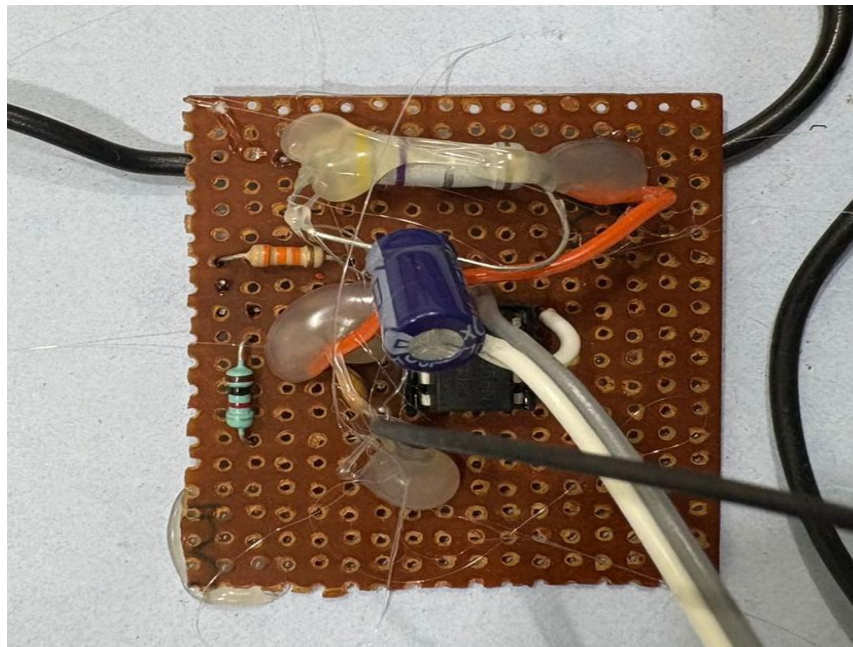
- **Voltage Sensor / Voltage Divider:**

- The circuit scales down a high input voltage, such as battery voltage, into a lower output voltage.
- The divider is constructed using two resistors connected in series.
- A small capacitor is included in parallel with Resistor to act as a filter.



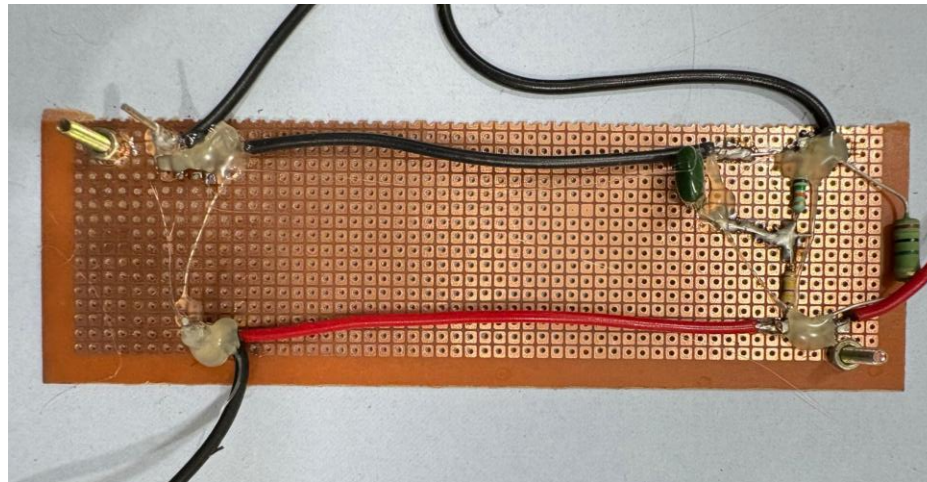
- **Current Sensor**

- The module measures the high electrical current flowing through the power wires without interrupting the circuit.
- The Op-Amp is responsible for amplifying the initial sensing signal and converting it into usable output voltage for the controller.
- It enables the microcontroller to actively monitor the current to prevent overcurrent conditions or short circuits.



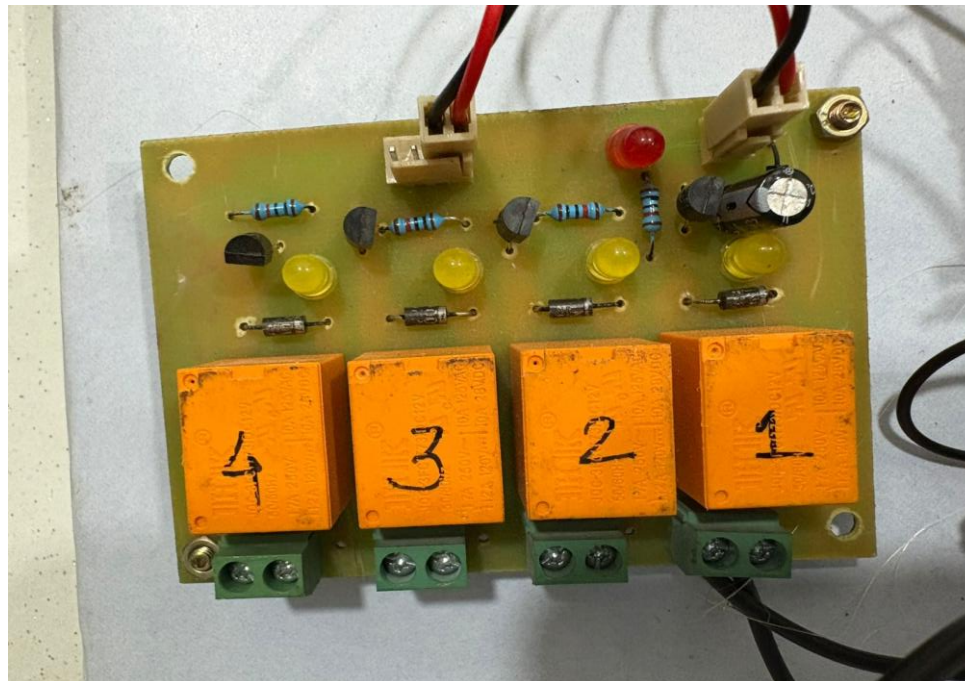
- **DC BUS**

- It acts as the central power supply that distributes electrical energy from the battery to all connected subsystems.
- It acts as an interface, connecting the energy source (the battery/charger) and DC- to - AC converter (the inverter board).
- The voltage on the DC Bus/DC Link is a critical parameter that is continuously measured and fed back to the microcontroller for control and protection.
- Controlling the stability and value of the DC Bus voltage is crucial for the correct and reliable operation



- **Relays**

- Type SPDT (Single-Pole, Double-Throw); 5 pins; Sugar Cube- PCB Mount; Voltage Rating: 12V & Current Rating: 7A.
- The relay features low coil power consumption, high current contacts, and a compact size; The relay is RoHS Compliant.



❖ Hardware Results :

- Grid is Active :

Sr.no.	Loading	Voltage	Current	Remark
1	No load	120V	0A	System healthy
2	1 bulb	114V	0.27A	System healthy
3	2 bulbs	113V	0.55A	System healthy
4	>2 bulbs	<110V	>0.7A	Overload detected and system shutdown

- UPS Active and grid is failed :

Sr.no.	Loading	Voltage	Current	Remark
1	No load	138V	0A	System healthy
2	1 bulb	123V	0.27A	System healthy
3	2 bulbs	122V	0.55A	System healthy
4	>2 bulbs	<118V	>0.7A	Overload detected and system shutdown

Conclusion

- A 120 V LVDC microgrid prototype was successfully designed and developed using grid and battery sources.
- The ESP32 controller effectively monitored voltage and current, ensuring real-time system supervision
- Automatic switching between grid and UPS was achieved through sugar cube relays, maintaining power continuity.
- The system provided reliable overload protection by detecting excess current and tripping the circuit.
- The system provided reliable overload protection by detecting excess current and tripping the circuit

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